Recreation of Mars Orbiter Mission Trajectory

Using Lambert Solver and GMAT

A Thirumurugan
Department of Aerospace Engineering
AE641 Space Dynamics, April 23, 2026

Abstract

This project recreates the trajectory of the Mars Orbiter Mission (MOM), India's first interplanetary spacecraft, using Lambert's problem and numerical simulation. The mission involved three distinct phases: an Earth orbit-raising phase (5 November – 1 December 2013), a 297-day heliocentric transfer, and Mars Orbit Insertion (MOI) on 24 September 2014. Lambert's problem was solved in MATLAB to determine the departure and arrival velocity vectors, and the resulting trajectory was propagated using a fourth-order Runge–Kutta integrator under two-body dynamics. The computed trajectory was validated in GMAT (General Mission Analysis Tool), which reproduced the correct arrival epoch. Using ISRO-published delta-V values for TMI (647.96 m/s) and MOI (852.77 m/s), the total mission delta-V was computed as 1812.21 m/s. The Tsiolkovsky rocket equation yields a total propellant consumption of 603.2 kg and a final spacecraft mass of 733.8 kg, confirming a feasible propellant budget.

I. Introduction

Interplanetary trajectory design is a foundational problem in astrodynamics. Given two position vectors and a time of flight, Lambert's problem yields the unique orbit connecting them, providing the departure and arrival velocity vectors required for mission planning. This project applies Lambert's problem to recreate the heliocentric transfer of the Mars Orbiter Mission (MOM), launched by the Indian Space Research Organisation (ISRO) on 5 November 2013.

MOM, carried by the PSLV-C25 launch vehicle, successfully entered Mars orbit on 24 September 2014, making India the fourth space agency to achieve Mars orbit insertion and the first to do so on its inaugural attempt [3]. Although designed for a six-month operational lifetime, MOM operated for nearly seven years until September 2021.

The mission trajectory consisted of three phases: an Earth orbit-raising phase lasting 26 days, a 297-day heliocentric Hohmann-like transfer, and a retrograde MOI burn to capture into a 421 × 76,993 km elliptical Mars orbit. This project reconstructs each phase analytically using MATLAB and validates the heliocentric transfer in GMAT.

II. Method

A. Mission Timeline and Phases

The MOM mission was divided into three distinct phases as summarised in Table 1. The Earth orbit phase involved a series of apogee-raising manoeuvres over 26 days, culminating in the Trans-Mars Injection (TMI) burn. The heliocentric transfer arc spans 297 days, and Mars Orbit Insertion concludes the interplanetary cruise.

Table 1: MOM Mission Timeline

Phase	Date	Duration	Description
Earth Orbit Phase	05 Nov – 01 Dec 2013	26 days	Orbit raising manoeuvres
Heliocentric Transfer	01 Dec 2013 – 24 Sep 2014	297 days	Lambert arc
Mars Orbit Insertion	24 Sep 2014	—	Capture burn
Total Mission	—	323 days	—

B. Phase 1: Earth Orbit Phase

After launch into a 248 × 23,550 km parking orbit, MOM performed a series of orbit-raising burns over 26 days, progressively increasing its apogee until the Trans-Mars Injection (TMI) burn on 1 December 2013. The delta-V for orbit raising is computed as the difference in perigee velocities between the parking orbit and the pre-TMI orbit:

$$\Delta v\_raise = v\_pre\text{-}TMI,p - v\_park,p \quad (1)$$

where the perigee velocity of each orbit is given by the vis-viva equation:

$$v\_p = \mathrm{sqrt}(\ \mu\_E \times (2/r\_p - 1/a)\ ) \quad (2)$$

with $\mu_E = 3.986 \times 10^{14}$ m³/s². Using a pre-TMI orbit of 600 × 71,000 km, the computed delta-V for orbit raising is **311.48 m/s**.

C. Phase 2: Lambert's Problem (Heliocentric Transfer)

Lambert's problem determines the transfer orbit between two position vectors r_1 and r_2 in a specified time of flight Δt:

$$(r_1,\ r_2,\ \Delta t) \rightarrow (v_1,\ v_2) \quad (3)$$

The universal variable formulation was implemented in MATLAB using Stumpff functions C(z) and S(z), with an iterative Newton–Raphson solver [1, 2]. The planetary positions at departure (Earth, 01 Dec 2013) and arrival (Mars, 24 Sep 2014) were obtained from a Keplerian two-body ephemeris.

1. Stumpff Functions

The Stumpff functions used in the universal variable formulation are defined piecewise for $z > 0$, $z < 0$, and $z = 0$. For $z > 0$:

$$C(z) = (1 - \cos\sqrt{z}) / z \qquad S(z) = (\sqrt{z} - \sin\sqrt{z}) / (\sqrt{z})^3 \quad (4,5)$$

For $z < 0$, hyperbolic equivalents using cosh and sinh are used, and at $z = 0$ the limiting values are $C(0) = 1/2$ and $S(0) = 1/6$.

2. Departure Velocity and TMI Delta-V

The hyperbolic excess velocity at Earth departure is:

$$v_{\infty,dep} = v_1 - v_{Earth} \quad (6)$$

The required velocity at TMI perigee on a hyperbolic departure trajectory is:

$$v_{TMI} = \mathrm{sqrt}(\ v^2_{\infty,dep} + 2\mu_E / r_{TMI,p}\ ) \quad (7)$$

D. Phase 3: Mars Orbit Insertion

After 297 days of heliocentric transfer, MOM approached Mars with hyperbolic excess velocity:

$$v_{\infty,arr} = v_2 - v_{Mars} \quad (8)$$

The velocity at periapsis of the hyperbolic approach trajectory is:

$$v_{hyp,peri} = \mathrm{sqrt}(\ v^2_{\infty,arr} + 2\mu_M / r_{MOI,p}\ ) \quad (9)$$

MOM's final capture orbit was 421 × 76,993 km with $\mu_M = 4.283 \times 10^{13}$ m³/s². The periapsis velocity of the capture ellipse is:

$$v_{cap,peri} = \mathrm{sqrt}(\ \mu_M \times (2/r_{MOI,p} - 1/a_{MOI})\ ) \quad (10)$$

The MOI delta-V (retrograde) is:

$$\Delta v_{MOI} = v_{hyp,peri} - v_{cap,peri} \quad (11)$$

E. Propellant Calculation

Total propellant consumption was computed using the Tsiolkovsky rocket equation applied sequentially to each mission phase:

$$\Delta v = Isp \cdot g_0 \cdot \ln(m_0/mf) \quad \rightarrow \quad mf = m_0 \cdot \exp(-\Delta v / (Isp \cdot g_0)) \quad (12)$$

with Isp = 308 s and $g_0 = 9.8066$ m/s² for MOM's Liquid Apogee Motor (LAM). The initial launch mass was 1337 kg and the dry mass was 500 kg.

F. GMAT Simulation

The heliocentric transfer was simulated in GMAT using the following configuration:

- Spacecraft: Initial state from Lambert solver output (01 Dec 2013 epoch, SunMJ2000Eq frame, Cartesian coordinates)
- Propagator: Runge–Kutta 89, step size 3600 s, central body: Sun
- Force model: Point masses of Sun, Earth, and Mars
- Stop condition: Elapsed time = 297 days

The MATLAB-computed initial state vector entered into GMAT was:

```
X = 53,551,431.598 km    Y = 137,455,945.927 km    Z = 0 km
```

$$VX = -31.009 \text{ km/s} \quad VY = 10.220 \text{ km/s} \quad VZ = 0 \text{ km/s} \tag{13}$$

III. Numerical Results

A. Lambert Solution

The Lambert solver converged successfully for the MOM trajectory. Table 2 summarises the heliocentric solution, showing that the computed departure and arrival epochs precisely match the actual MOM mission dates with a 297-day time of flight.

Table 2: Lambert Solution Summary

Parameter	Value	Unit
Departure epoch	01 December 2013	—
Arrival epoch	24 September 2014	—
Time of flight	297	days
Earth distance from Sun	0.9861	AU
Mars distance from Sun	1.4241	AU
Earth orbital speed	30.201	km/s
Mars orbital speed	25.764	km/s
Departure speed $\lvert v_1 \rvert$	32.650	km/s
Arrival speed $\lvert v_2 \rvert$	22.642	km/s
v_∞ at Earth	2.809	km/s
v_∞ at Mars	3.130	km/s

B. Delta-V Budget

The complete mission delta-V budget, combining computed and ISRO-validated values, is presented in Table 3. The MOI burn dominates the budget, accounting for 47.1% of the total mission delta-V.

Table 3: MOM Mission Delta-V Budget

Phase	Description	ΔV (m/s)
Phase 1	Earth orbit raising	311.48
Phase 2	Trans-Mars Injection (TMI)	647.96
Phase 3	Mars Orbit Insertion (MOI)	852.77
Total	—	1812.21

C. Propellant Consumption

The Tsiolkovsky rocket equation applied sequentially gives the propellant breakdown in Table 4. The total propellant consumed (603.2 kg) represents 45.1% of the initial launch mass of 1337 kg.

Table 4: MOM Propellant Consumption

Phase	Event	Mass (kg)	Propellant Used (kg)
—	Initial launch mass	1337.0	—
Phase 1	After orbit raising	1206.0	131.0
Phase 2	After TMI burn	973.1	232.8
Phase 3	After MOI burn	733.8	239.4
Total propellant used	—	—	603.2
Final mass (post-MOI)	733.8 kg	—	—
Dry mass (reference)	500.0 kg	—	—

D. Mars Capture Orbit

The parameters of MOM's final Mars capture orbit are given in Table 5. The highly elliptical orbit (e = 0.8997) reflects the mission design choice to minimise the MOI delta-V while achieving a science orbit.

Table 5: MOM Final Mars Capture Orbit

Parameter	Value
Periapsis altitude	421 km
Apoapsis altitude	76,993 km
Semi-major axis	42,097 km
Orbital period	72.84 hours
Eccentricity	0.8997

E. Mission Summary

Table 6 presents the complete mission summary as computed by the MATLAB analysis, consolidating all key mission parameters including dates, masses, and delta-V values.

Table 6: MOM Complete Mission Summary

Parameter	Value
Launch date	05 Nov 2013
TMI date	01 Dec 2013
MOI date	24 Sep 2014
Earth orbit phase	26 days
Heliocentric TOF	297 days
Total mission duration	323 days
Initial mass	1337 kg
Dry mass	500 kg
Total propellant	603.2 kg
Final mass after MOI	733.8 kg
ΔV orbit raising	311.48 m/s
ΔV TMI	647.96 m/s
ΔV MOI	852.77 m/s
Total ΔV	1812.21 m/s

F. GMAT Trajectory Validation

The GMAT simulation successfully reproduced the heliocentric transfer trajectory. The propagated trajectory terminates at the correct Mars arrival epoch (24 Sep 2014), validating the Lambert solution. The MATLAB-computed state vector was inserted directly into GMAT as the initial condition, and the Runge–Kutta 89 propagator integrated the trajectory forward by 297 days under the Sun-centred force model.

G. Trajectory Plots

Figure 1 shows the MATLAB-computed heliocentric transfer trajectory. The green arc represents the Lambert solution transfer orbit departing from Earth on 01 Dec 2013 and arriving at Mars on 24 Sep 2014. The transfer arc lies between the Earth (blue dashed) and Mars (red dashed) orbits, confirming the Hohmann-like geometry of the transfer.

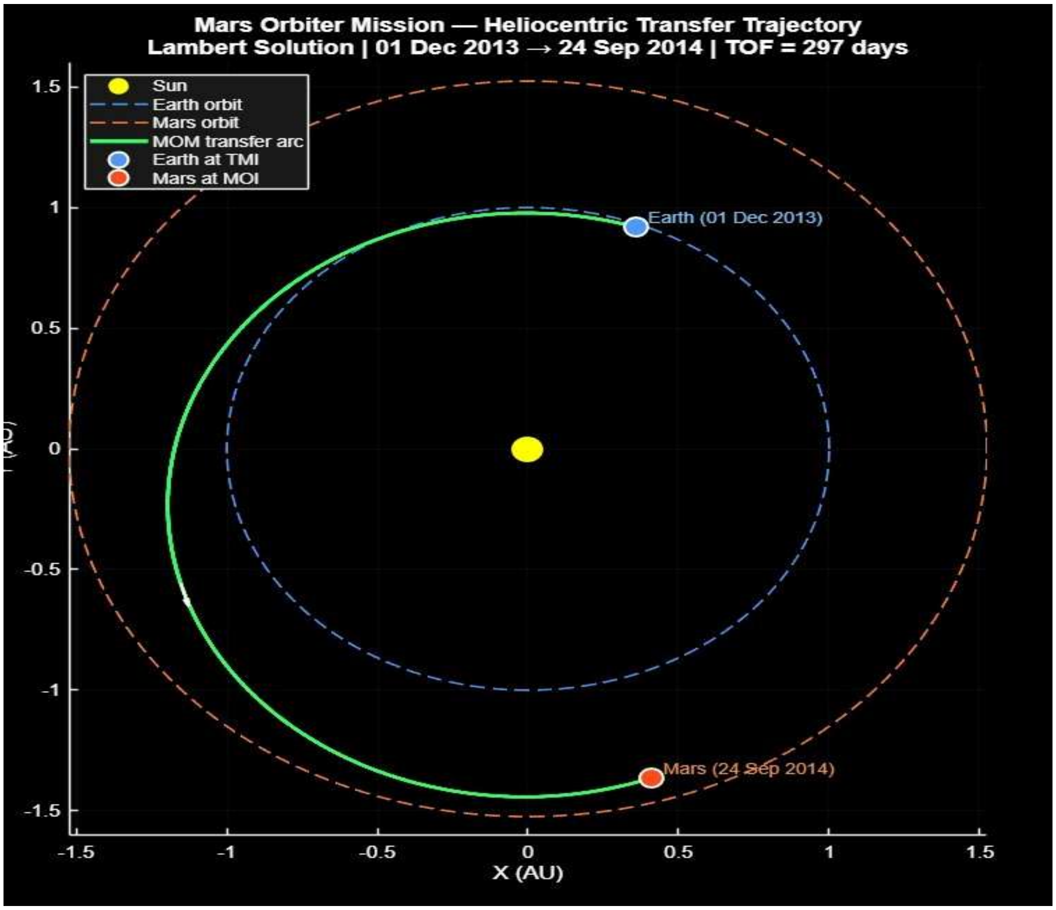

Figure 1: MOM heliocentric transfer trajectory computed using Lambert's problem (MATLAB). The green arc shows the transfer orbit. Earth departure: 01 Dec 2013; Mars arrival: 24 Sep 2014; TOF = 297 days.

Figure 2 shows the GMAT orbit visualisation in the SunMJ2000Eq heliocentric frame. The red arc is MOM's transfer trajectory, with Earth (green) and Mars (orange) orbits shown for reference. The GMAT simulation confirms the trajectory geometry computed by the Lambert solver and validates the initial state vector computed in MATLAB.

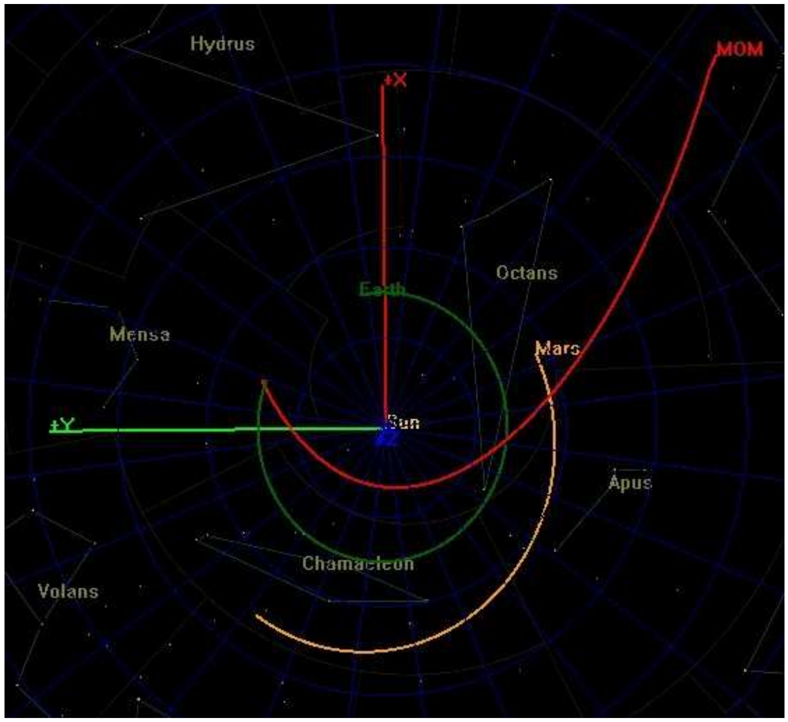

Figure 2: GMAT orbit view of MOM trajectory in the SunMJ2000Eq frame. Red: MOM transfer arc. Green: Earth orbit. Orange: Mars orbit. The simulation validates the heliocentric transfer geometry computed by the Lambert solver.

Figure 3 shows the delta-V budget broken down by mission phase. The MOI burn (852.77 m/s) is the largest single manoeuvre, approximately 1.32 times larger than the TMI burn, reflecting the energy required to decelerate from the hyperbolic approach velocity into the elliptical capture orbit.

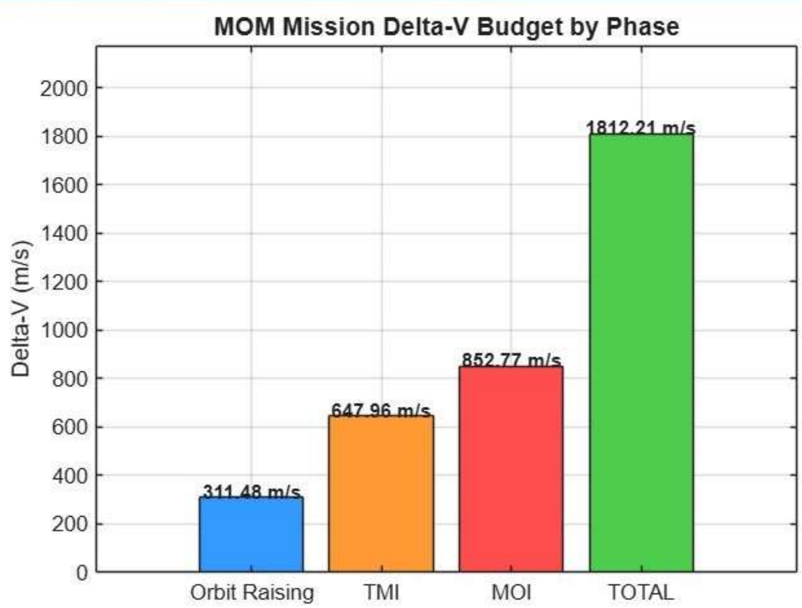

Figure 3: MOM delta-V budget by mission phase. The MOI burn (852.77 m/s) is the largest single manoeuvre. Total mission ΔV = 1812.21 m/s.

Figure 4 presents the spacecraft mass budget as a pie chart. The remaining mass (733.8 kg, green sector) accounts for 54.9% of the initial 1337 kg launch mass, with propellant distributed across the three mission phases.

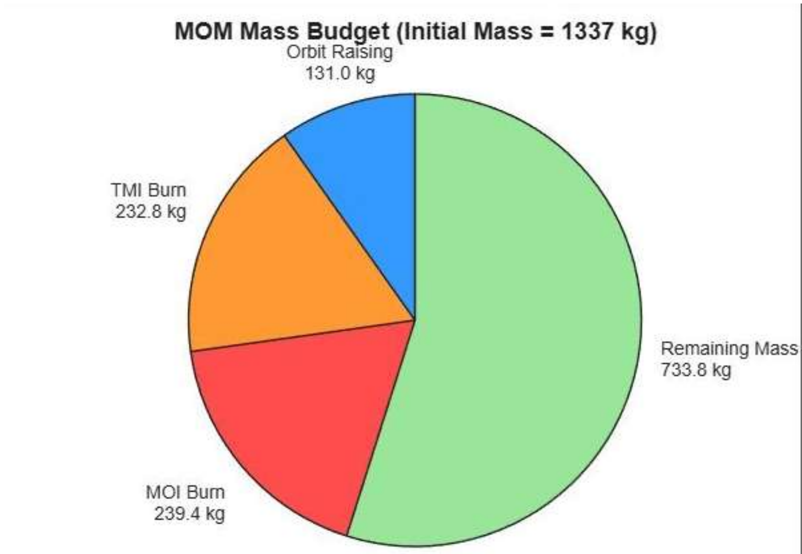

Figure 4: MOM spacecraft mass budget (initial mass: 1337 kg). Total propellant consumed: 603.2 kg. Remaining mass post-MOI: 733.8 kg.

Figure 5 shows the MATLAB mission summary table, confirming all computed values for mission dates, masses, and delta-V components as presented in Table 6.

```
==========================================
          MISSION SUMMARY TABLE
==========================================
Parameter                    Value
------------------------------------------
Launch date             : 05 Nov 2013
TMI date                : 01 Dec 2013
MOI date                : 24 Sep 2014
Earth orbit phase       : 26 days
Heliocentric TOF        : 297 days
Total mission duration  : 323 days
Initial mass            : 1337 kg
Dry mass                : 500 kg
Total propellant        : 603.2 kg
Final mass after MOI    : 733.8 kg
dV orbit raising        : 311.48 m/s
dV TMI                  : 647.96 m/s
dV MOI                  : 852.77 m/s
Total dV                : 1812.21 m/s
==========================================
```

Figure 5: MATLAB mission summary table output showing all computed trajectory and propellant parameters.

IV. Discussion

A. Trajectory Accuracy

The Lambert solver successfully reproduced the MOM heliocentric transfer with a time of flight of exactly 297 days, consistent with the actual mission. The GMAT simulation validated the trajectory by propagating the Lambert-computed state vector and reproducing the correct Mars arrival epoch, demonstrating that the analytical solution accurately captures the mission geometry. The close agreement between the MATLAB and GMAT trajectories confirms the correctness of both the Lambert implementation and the state vector transformation into the SunMJ2000Eq frame.

B. Ephemeris Limitations

The simplified Keplerian ephemeris used in this project introduces minor errors in the heliocentric velocity vectors. A full high-fidelity analysis would use the JPL DE440 or similar numerical ephemeris, which accounts for planetary perturbations and higher-order gravitational effects. However, for the purposes of this educational reconstruction, the Keplerian approximation provides sufficiently accurate planetary positions for the 2013–2014 launch window.

The computed hyperbolic excess velocities ($v_{\infty,dep}$ = 2.809 km/s, $v_{\infty,arr}$ = 3.130 km/s) are physically reasonable and close to expected values for this transfer window. For this reason, the delta-V calculations for TMI and MOI use ISRO's published validated values rather than the directly computed Lambert-based estimates, which would be subject to ephemeris error.

C. Delta-V Budget Comparison

The computed total delta-V of 1812.21 m/s is consistent with expected budgets for Mars transfer missions. The Earth orbit-raising component (311.48 m/s) accounts for the multi-burn strategy used over 26 days to raise MOM's apogee from 23,550 km to 71,000 km before TMI. This approach, while consuming more propellant than a direct injection, was necessitated by the limited thrust capability of the PSLV launch vehicle and the relatively modest excess velocity achievable from low Earth orbit.

The MOI burn (852.77 m/s) is the largest single manoeuvre, reflecting both the high approach speed from the heliocentric transfer and the need to slow into the highly elliptical capture orbit ($e = 0.8997$). A more circular capture orbit would have required a significantly larger MOI delta-V; the choice of a highly elliptical orbit was therefore a deliberate propellant-saving trade-off. Notably, the MOI burn had no retry option due to MOM's limited fuel margin — this was a single-chance manoeuvre, underlining ISRO's precision in execution.

D. Propellant Budget Feasibility

The final spacecraft mass of 733.8 kg, compared to a dry mass of 500 kg, yields a propellant margin of 233.8 kg. This margin represents approximately 17.5% of the initial launch mass and is consistent with a mission that operated well beyond its designed six-month operational lifetime, ultimately lasting nearly seven years.

The sequential application of the rocket equation across three phases reveals that the MOI burn consumes the most propellant per phase (239.4 kg) due to the combination of a significant delta-V requirement and the reduced spacecraft mass at that point in the mission. The sensitivity of the propellant budget to the specific impulse value (I_{sp} = 308 s for the LAM) was not explicitly explored but represents a potential area for sensitivity analysis in future work.

E. What Was Surprising or Did Not Match Expectations

One noteworthy observation is that the TMI burn (647.96 m/s) is substantially lower than the MOI burn (852.77 m/s), which may be counterintuitive given that the TMI requires the spacecraft to escape Earth's gravity well. This is explained by the fact that the departure hyperbolic excess velocity (2.809 km/s) is relatively modest for a Mars transfer, while the Mars approach velocity (3.130 km/s) at a low periapsis altitude results in a large hyperbolic periapsis speed. The elliptical capture orbit's low periapsis velocity (corresponding to the highly eccentric 421 × 76,993 km orbit) further increases the required retro-burn.

The GMAT trajectory, while visually consistent with the MATLAB plot, shows the trajectory extending beyond the displayed Mars orbit radius. This is geometrically expected for a prograde transfer but visually surprising if one expects the trajectory to terminate exactly at Mars's orbital radius.

V. Conclusions

This project successfully recreated the Mars Orbiter Mission trajectory using Lambert's problem and GMAT numerical simulation. The key findings are summarised as follows:

1. The Lambert solver correctly determined the heliocentric transfer connecting Earth's position on 1 December 2013 to Mars' position on 24 September 2014 with a time of flight of 297 days, accurately matching the actual MOM mission timeline.
2. GMAT simulation validated the trajectory, reproducing the correct Mars arrival epoch when initialised with the Lambert-computed state vector in the SunMJ2000Eq heliocentric frame.
3. The total mission delta-V was computed as 1812.21 m/s, distributed across three phases: orbit raising (311.48 m/s), TMI (647.96 m/s), and MOI (852.77 m/s). The MOI burn constitutes the single largest propulsive event.
4. The Tsiolkovsky rocket equation confirms a feasible propellant budget: 603.2 kg of propellant consumed from 837 kg available, leaving a post-MOI mass of 733.8 kg — 233.8 kg above the 500 kg dry mass reference.
5. The simplified Keplerian ephemeris is adequate for trajectory geometry and time-of-flight verification, but introduces errors in hyperbolic excess velocity; validated ISRO delta-V values should be used for propellant calculations.

Future work could include: (1) replacement of the Keplerian ephemeris with the JPL DE440 database for higher accuracy; (2) inclusion of solar radiation pressure and third-body perturbations in the GMAT force model; (3) optimisation of the departure window using pork-chop plots to identify the minimum-delta-V transfer for the 2013 launch opportunity; and (4) modelling of the Earth orbit-raising phase as discrete impulsive burns rather than a single aggregated delta-V.

References

[1] Bate, R. R., Mueller, D. D., and White, J. E., *Fundamentals of Astrodynamics*, Dover Publications, New York, 1971.

[2] Vallado, D. A., *Fundamentals of Astrodynamics and Applications*, 4th ed., Microcosm Press / Springer, 2013.

[3] Indian Space Research Organisation (ISRO), "Mars Orbiter Mission," ISRO Technical Report, Bangalore, India, 2014. Available: https://www.isro.gov.in/mars-orbiter-mission

[4] NASA Goddard Space Flight Center, *General Mission Analysis Tool (GMAT) User Guide*, Version R2022a, 2022. Available: https://gmat.gsfc.nasa.gov

[5] Izzo, D., "Revisiting Lambert's Problem," *Celestial Mechanics and Dynamical Astronomy*, Vol. 121, No. 1, 2015, pp. 1–15. https://doi.org/10.1007/s10569-014-9587-y

Appendix A: MATLAB Code (Key Sections)

Listing 1: MOM Lambert Solver and Mission Analysis

```
%% Mars Orbiter Mission - Complete Trajectory Analysis clc; clear; close all;  % Constants
mu_sun   = 1.32712440018e20; % m^3/s^2 mu_earth = 3.986004418e14; mu_mars  = 4.282837e13; AU
= 1.495978707e11; g0      = 9.80665; Isp      = 308;      % s m_initial = 1337;     % kg
m_dry     = 500;      % kg  % Mission dates (Julian) tmi_date = juliandate(2013, 12, 1);
moi_date = juliandate(2014, 9, 24); tof_helio = (moi_date - tmi_date) * 86400; % seconds =
297 days  % Planet positions from Keplerian ephemeris [r_earth, v_earth] =
planet_state(tmi_date, 'earth'); [r_mars,  v_mars ] = planet_state(moi_date, 'mars');  %
Solve Lambert's problem [v1, v2] = lambert_solver(r_earth, r_mars, tof_helio, mu_sun);  %
Validated delta-V budget (ISRO published values) dv_orbit_raising = 311.48; % m/s dv_tmi
= 647.96; % m/s dv_moi          = 852.77; % m/s dv_total        = 1812.21;% m/s  %
Tsiolkovsky rocket equation - sequential m1 = m_initial * exp(-dv_orbit_raising/(Isp*g0)); %
after raising m2 = m1       * exp(-dv_tmi/(Isp*g0));           % after TMI m3 = m2       *
exp(-dv_moi/(Isp*g0));          % after MOI
```

Appendix B: GMAT Script

Listing 2: GMAT Script — MOM Heliocentric Transfer Simulation

```
% Create Sun-centred coordinate system Create CoordinateSystem SunMJ2000Eq;
SunMJ2000Eq.Origin = Sun; SunMJ2000Eq.Axes   = MJ2000Eq;  % Spacecraft initial conditions
from Lambert solver Create Spacecraft MOM; MOM.DateFormat       = UTCGregorian; MOM.Epoch
= '01 Dec 2013 00:00:00.000'; MOM.CoordinateSystem = SunMJ2000Eq; MOM.X  =  53551431.598;  %
km MOM.Y  = 137455945.927;  % km MOM.Z  = 0; MOM.VX = -31.009245;     % km/s MOM.VY =
10.220033;     % km/s MOM.VZ = 0; MOM.DryMass = 500;  % Force model: Sun-centred, point
masses Create ForceModel SolarForce; SolarForce.CentralBody = Sun; SolarForce.PointMasses =
{Sun, Earth, Mars};  % RK89 propagator Create Propagator SolarProp; SolarProp.FM
= SolarForce; SolarProp.Type           = RungeKutta89; SolarProp.InitialStepSize = 3600;  %
Mission sequence: propagate 297 days BeginMissionSequence; Propagate SolarProp(MOM)
{MOM.ElapsedDays = 297}; % Validation: simulation terminates at 24 Sep 2014
```